

HUSEYIN KOCAKUSAK

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PROFESSIONAL EXPERIENCE

Researcher; Genç Neurodegeneration Laboratory

July 2023 - January 2026

- Investigating the protective effects of IL-4 induced Microglia derived small Extracellular vesicles against Rotenone-induced toxicity on SH-SY5Y cells and Zebrafish.
- Analyzing **small RNA sequencing** data by leveraging the power of Linux, Python and R to uncover crucial insights into the role of **tRNA-derived fragments and miRNAs** in Neuroinflammation & Neurodegeneration.
- Conducting **neuroinflammatory and neuroprotective applications** to observe outcomes by using SH-SY5Y, HMC3, N2A and N9 cell lines in **cell culture**.
- Performed **behavioral tests, brain extraction and maintenance** of Neat1 KO & NLRP3 mice and rats in **Vivarium** 
- Microplastics and Nanoplastics (MNPs) role in neurodegeneration.
- Awarded and supported** by BİDEB-2210/A MSc scholarship programme by TÜBİTAK

Student Researcher; Akgül's ncRNA Laboratory

July 2021 - June 2023

- Project Executive** in a bioinformatics analysis project named as “long non-coding RNAs affected in conditions of Cisplatin and Doxorubicin associated apoptosis.” within the scope of 2209/A, TÜBİTAK.
- Designed primers** for YTH domain proteins on Snapgene, PCR reactions, bacterial transformations and **Western Blotting** to observe **post-transcriptional modifications**.
- Performed TRIzol **RNA Isolations** (HEK293, HeLa, & MCF7), **RT-qPCR analysis** & **Western Blot Analysis** to assist experiments of Master's & Doctorate students.
- Awarded and supported** by BİDEB-2209/A & STAR scholarship programmes by TÜBİTAK.

Summer Internship; Genç Neurodegeneration Laboratory

July - September 2022

- Involved in obtaining **NEAT1 lncRNA knockout** mice experiments by performing PCR and gel electrophoresis.
- Conducted cell passages** of N9 microglia cell line to learn how to deal with cell culture and do other experiments such as cell viability tests throughout the internship.
- Performed **Cytotoxicity test** by determining the release of LDH from N9 microglia cells; findings were presented to improve the quality of basic experiments of laboratory.
- Performed **Immunofluorescence staining** by using NLRP3 and IBA1 antibodies and Fluorescent microscopy were used to acquire images. Afterwards, images were analyzed with ImageJ software.
- Assisted **microglia primary cell culture** procedure by only **chopping brains** with the help of a razor blade to be used in further experiments.

Voluntary Translator; Evrim Ağacı (Tree of Evolution)

July 2019 - April 2020

- Translated** and wrote different kinds of topics to spread science to everyone in Turkey. Got almost 700,000 reads with 15 content

ARTICLES

- Kocakusak H, Kök AB, Ozturk B, Karacicek B and Genc S (2025). Deciphering the role of tRNA-derived fragments in neurological and psychiatric disease pathogenesis.** Front. Cell. Neurosci. 19:1663788. doi: 10.3389/fncel.2025.1663788
- Arioz B. I., Binokay L., **KOCAKUSAK H.** Karakulah G., Genc S. (2026). **Profiling tRNA-derived fragments in LPS-induced microglia and their influence on immune response** . Brain Research, 150431, ISSN 0006-8993, <https://doi.org/10.1016/j.brainres.2026.150431>.
- H. Kocakusak**, C. P. Gonul, B. I. Arioz, L. Binokay, G. Karakulah, S. Genc. **Polarized Microglia Extracellular Vesicles has distinct tRNA-derived small RNA profiles.** (In Preparation)

CONGRESS ATTENDED

- 22nd Turkish Neuroscience Congress | Istanbul, Turkey, 2024
- (Poster) 1st International Extracellular Vesicles Conference | Ankara, Turkey, 2024
- Exposome-TR Symposium | Izmir, Turkey, 2024
- (Poster) The 8th Venusberg Meeting on Neuroinflammation | Belval, Luxemburg, May, 2025
- (Organizaton Team) 23rd Turkish Neuroscience Congress | Izmir, Turkey, 2025”
- 2nd International Extracellular Vesicles Conference | Ankara, Turkey, 2026

EDUCATION

Dokuz Eylül University, Izmir Biomedicine and Genome Center | Master of Science

Non-coding RNAs, Bioinformatics, Neurodegeneration, Neuroprotection | GPA: 3.93 / 4.00

Izmir Institute of Technology | Bachelor of Science | Izmir, Urla | August 2023

Molecular Biology, Non-coding RNAs, Apoptosis & Bioinformatics | GPA: 3.51 / 4.00

SKILLS & INTERESTS

Interests: **Beekeeping**, Fitness & Nutrition, Running, Camping, Mountaineering, Folk Dances

Languages: Native in Turkish & Fluent in English (IELTS 7.0), Beginner in German

Technical: Figma, Python, R, Ubuntu, HTML & CSS,